TUGAS BESAR ANTENA & PROPAGASI

SEMESTER 5

Kelompok 14

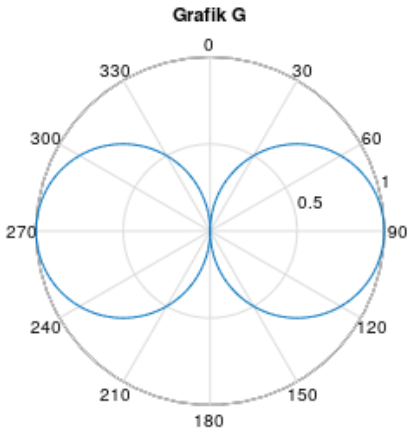
1. Elita Lestari Setianto (201344008)
2. M. Akmal Amanullah (201344021)

Kelas 3 NK

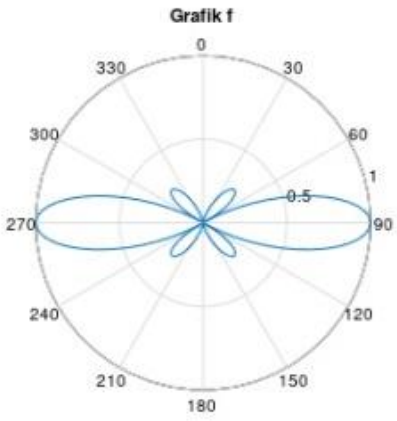
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1. Sebuah antena susun (array) *broadside* terdiri dari 4 buah elemen antena dipol ideal. Jika jarakantar elemen $d = \lambda/2$, tentukan dan plot pola elemen pola elemen $g(\theta, \phi)$, pola faktor susun
   $f(\theta, \phi)$ dan pola total $F(\theta, \phi)$ untuk masing-masing beda fasa antar elemen α:
   (a) α = 0°

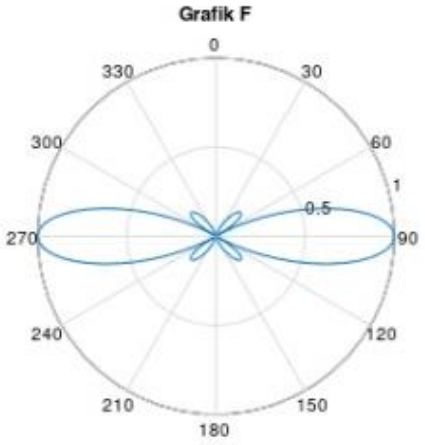
   - **Pola elemen g(θ,φ)**



   - **Pola factor susun f(θ,φ)**

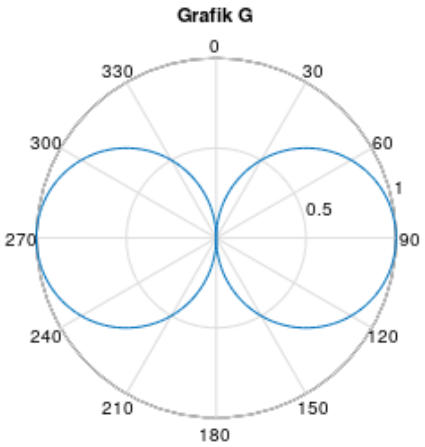


- **Pola total F(θ,φ)**

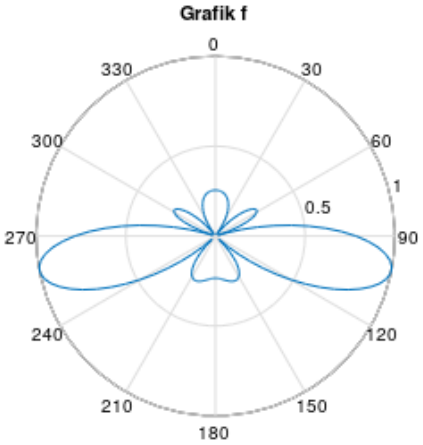


(b) α = 30°

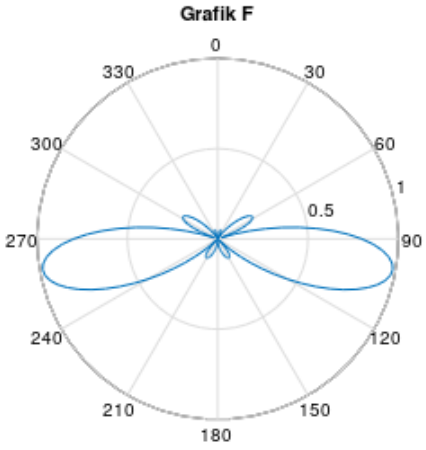
- **Pola elemen g(θ,φ)**



- **Pola factor susun f(θ,φ)**

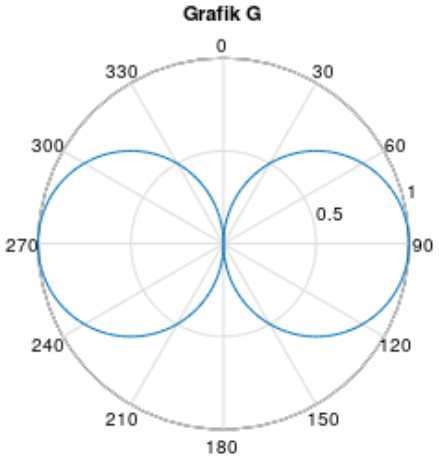


- **Pola total F(θ,φ)**

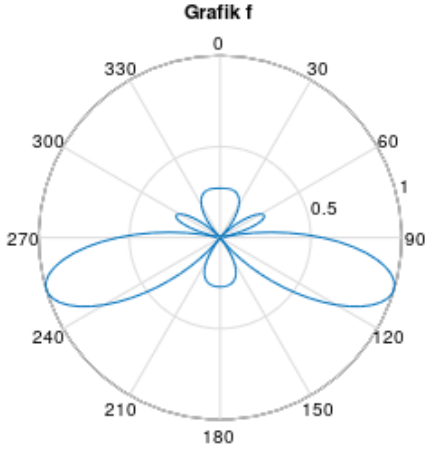


(c) α = 45°

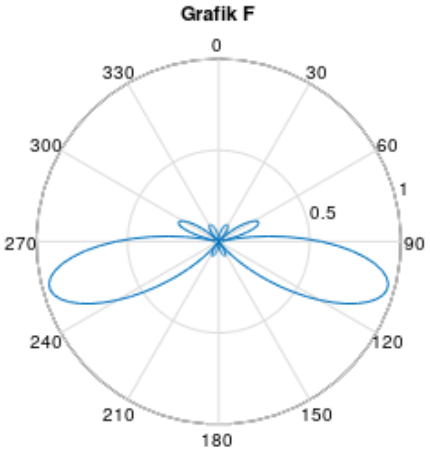
- **Pola elemen g(θ,φ)**



- **Pola factor susun f(θ,φ)**

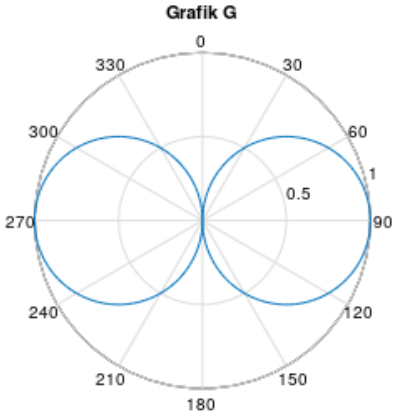


- **Pola total F(θ,φ)**

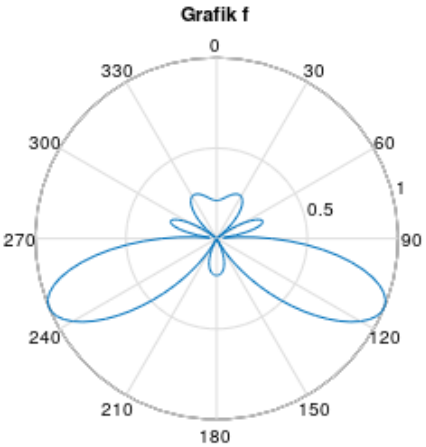


(d) α = 60°

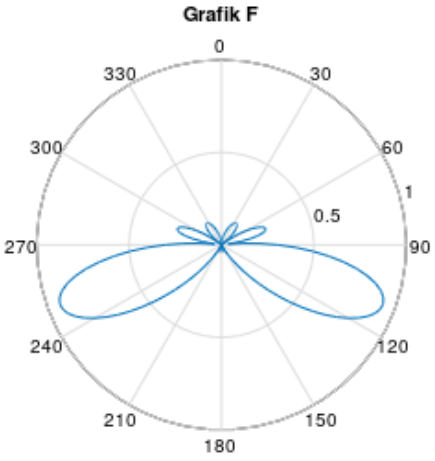
- **Pola elemen g(θ,φ)**



- **Pola factor susun f(θ,φ)**

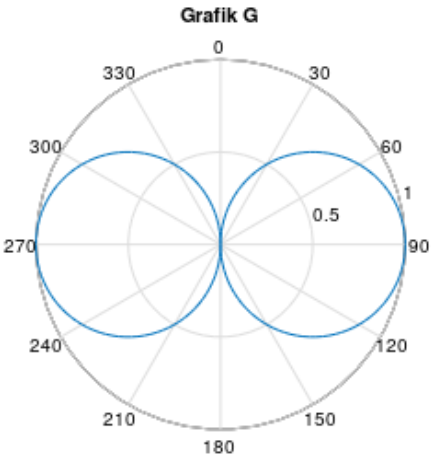


- **Pola total F(θ,φ)**

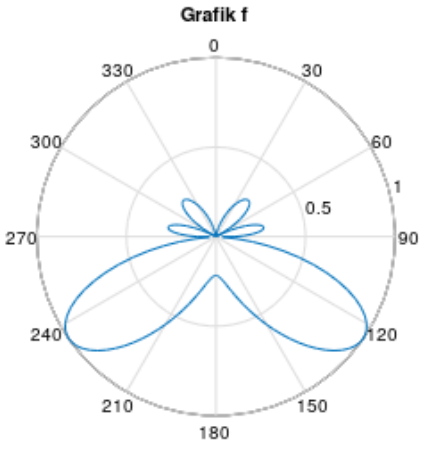


(e) α = 90°

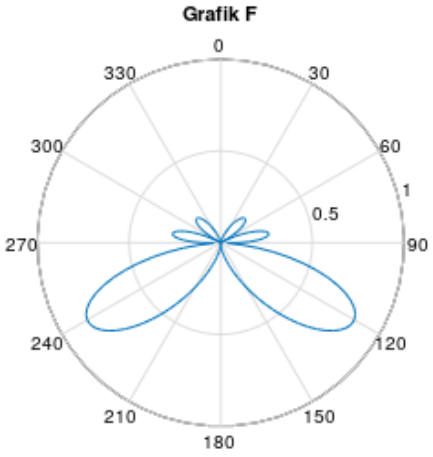
- **Pola elemen g(θ,φ)**



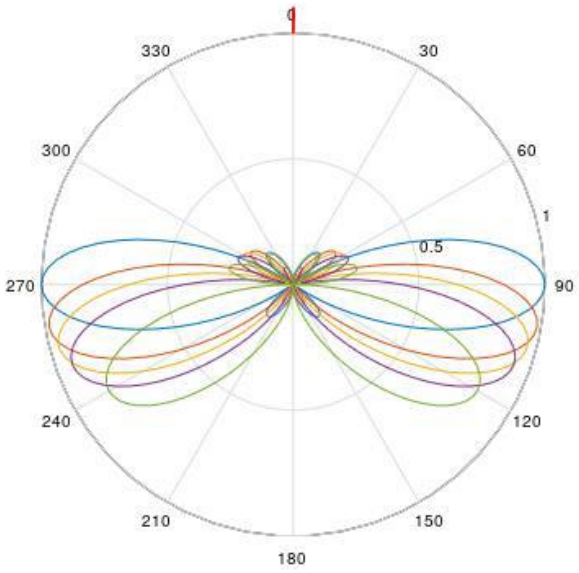
- **Pola factor susun f(θ,φ)**



- **Pola total F(θ,φ)**



- **Gambar total plot dalam satu grafik**

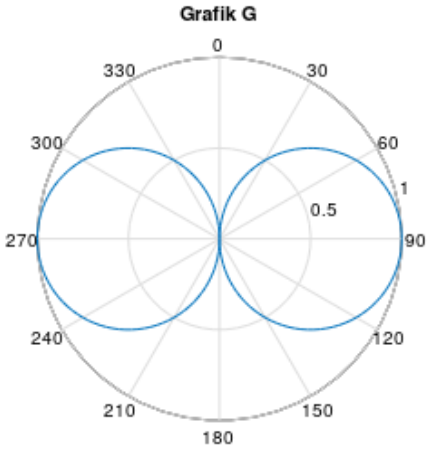


- (Biru) α = 0 °
- (Merah) α = 30 °
- (Kuning) α = 45 °
- (Ungu) α = 60 °
- (Hijau) α = 90 °

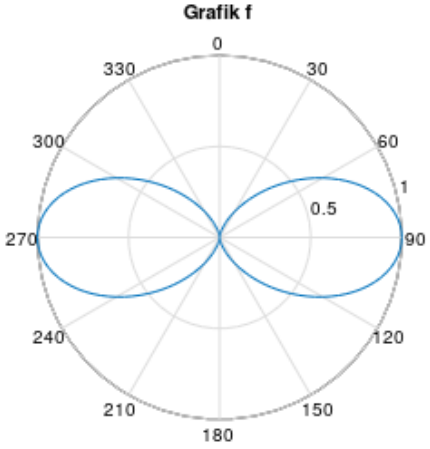
2. Sebuah antena susun (array) *broadside* terdiri dari 4 buah elemen antena dipol ideal. Jika bedafasa antar elemen α = 0°, tentukan dan plot pola elemen pola elemen $g(\theta, \phi)$, pola faktor susun $f(\theta, \phi)$ dan pola total $F(\theta, \phi)$ masing-masing untuk jarak antar elemen d:

   (a) $d = \lambda/4$

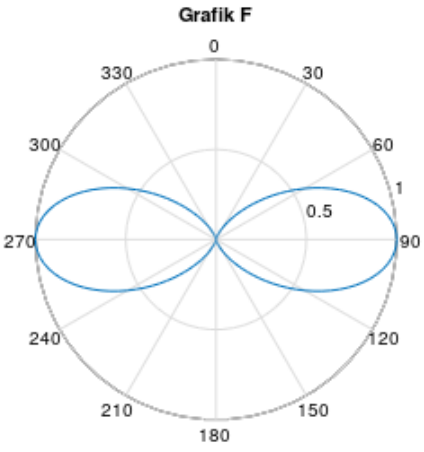
   - **Pola elemen g(θ,φ)**



- **Pola factor susun f(θ,φ)**

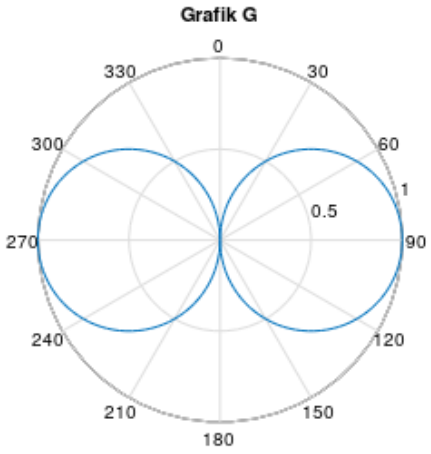


- **Pola total F(θ,φ)**

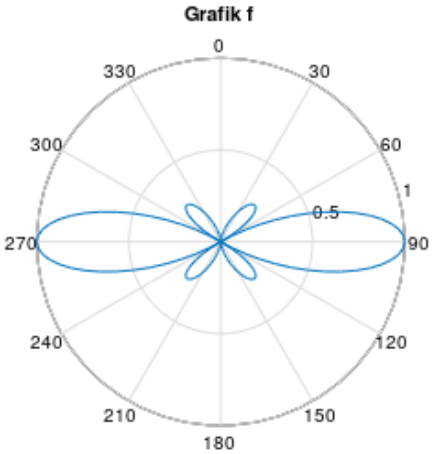


(b) $d = \lambda 2$

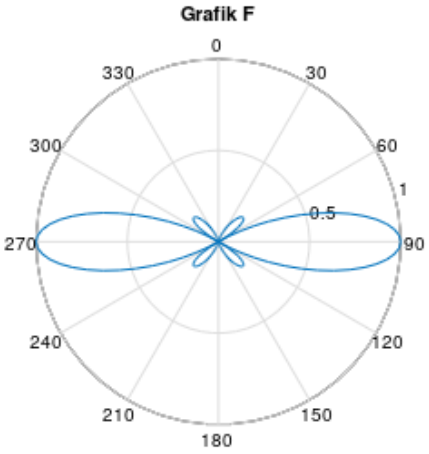
- **Pola elemen g(θ,φ)**



- **Pola factor susun f(θ,φ)**

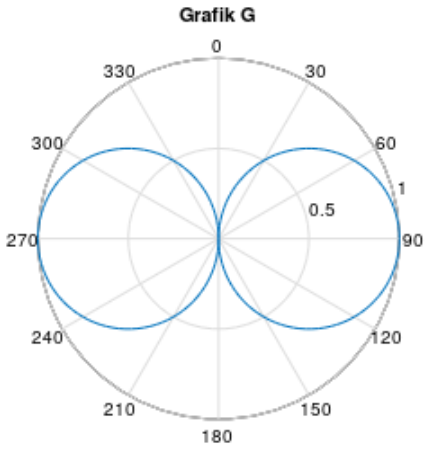


- **Pola total F(θ,φ)**

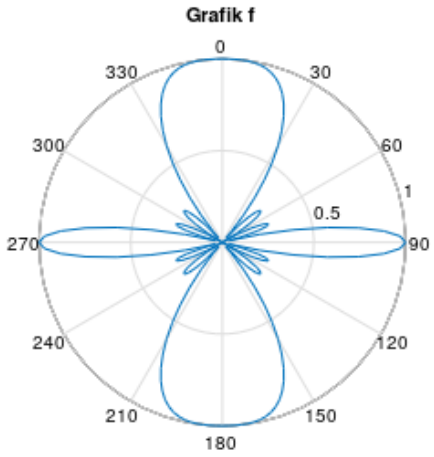


(c) $d = \lambda$

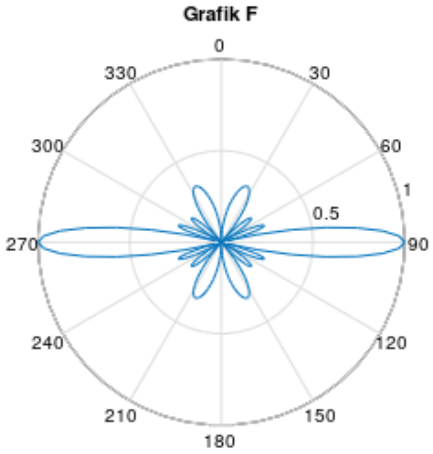
- **Pola elemen g(θ,φ)**



- **Pola factor susun f(θ,φ)**

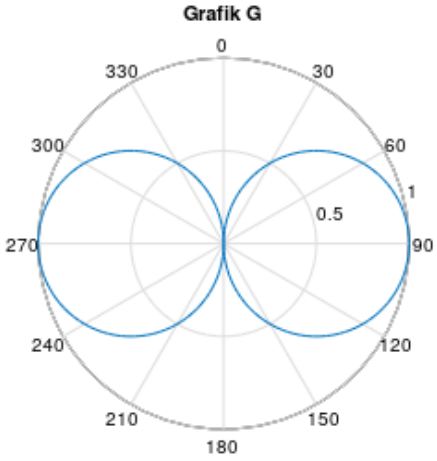


- **Pola total F(θ,φ)**

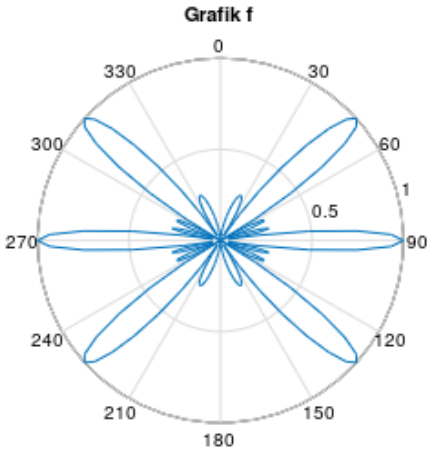


(d) $d = 3\lambda/2$

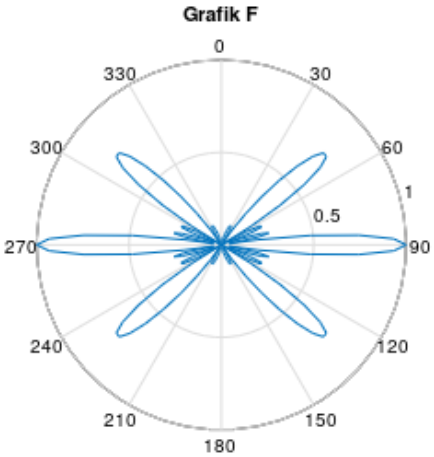
- **Pola elemen g(θ,φ)**



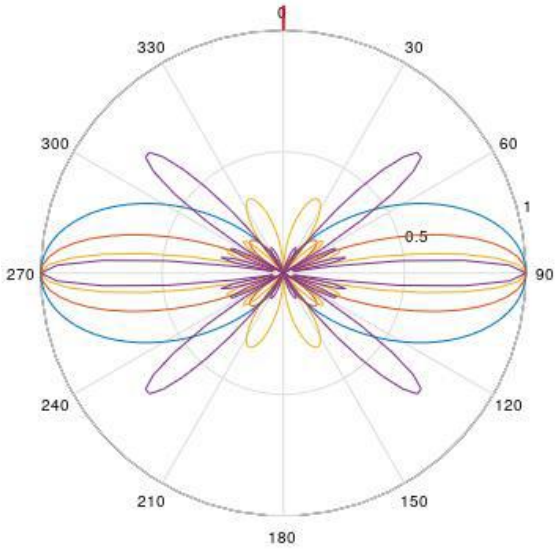
- **Pola factor susun f(θ,φ)**



- **Pola total F(θ,φ)**



- **Gambar total plot dalam satu grafik**

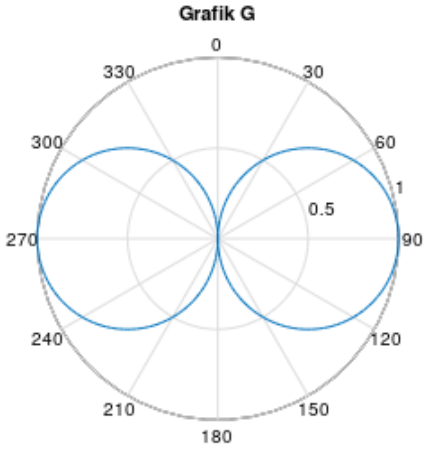


- (Blue) d = $\lambda/4$
- (Red) d= $\lambda/2$
- (Yellow) d = λ
- (Purple) d = $3\lambda/2$

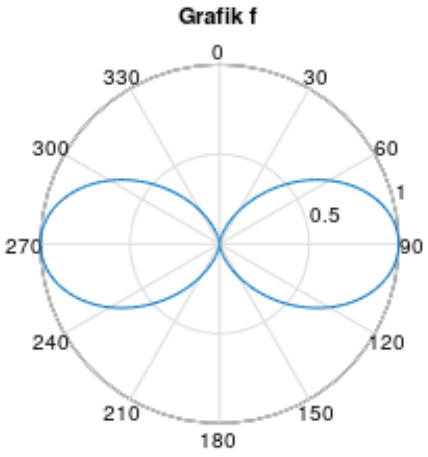
3. Sebuah antena susun (array) *broadside* terdiri dari n buah elemen antena dipol ideal. Jika bedafasa antar elemen α = 0°, dan jarak anter elemen $d = \lambda/2$, tentukan dan plot pola elemen pola elemen $g(\theta, \phi)$, pola faktor susun $f(\theta, \phi)$ dan pola total $F(\theta, \phi)$ masing-masing untuk jumlah elemen n:

   (a) n = 2

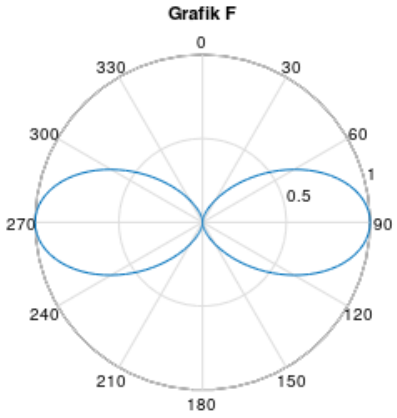
   - **Pola elemen g(θ,φ)**



- **Pola factor susun f(θ,φ)**

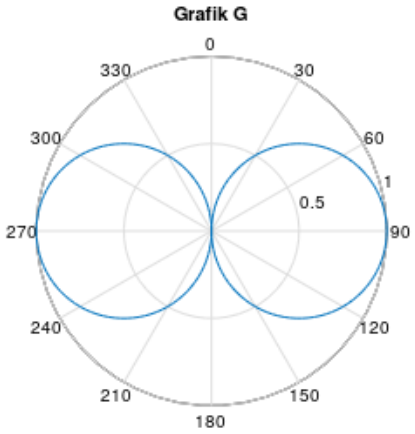


- **Pola total F(θ,φ)**

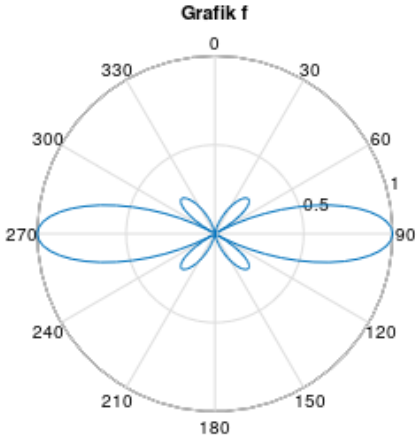


(b) n **= 4**

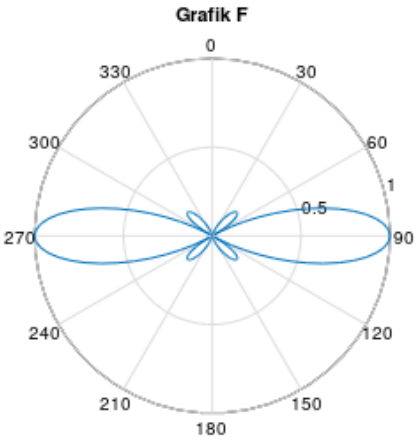
- **Pola elemen g(θ,φ)**



- **Pola factor susun f(θ,φ)**

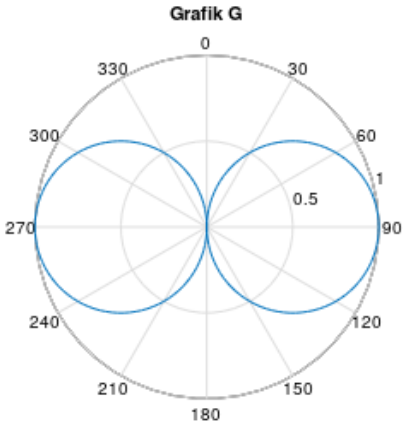


- **Pola total F(θ,φ)**

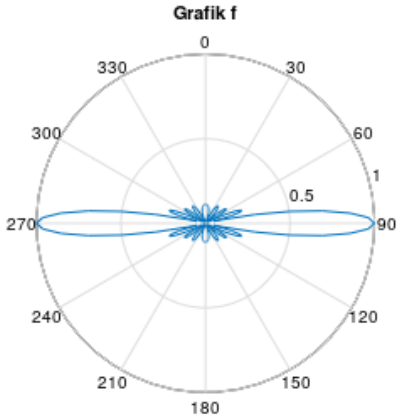


(c) n **= 8**

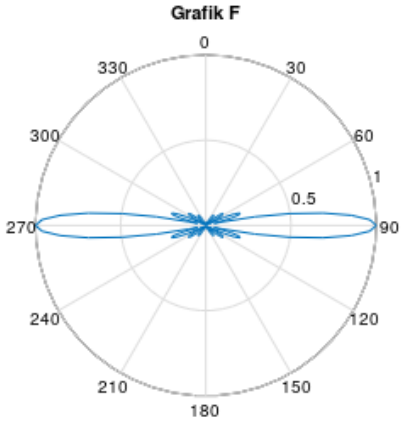
- **Pola elemen g(θ,φ)**



- **Pola factor susun f(θ,φ)**

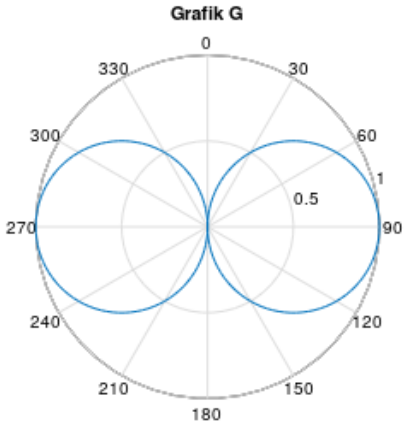


- **Pola total F(θ,φ)**

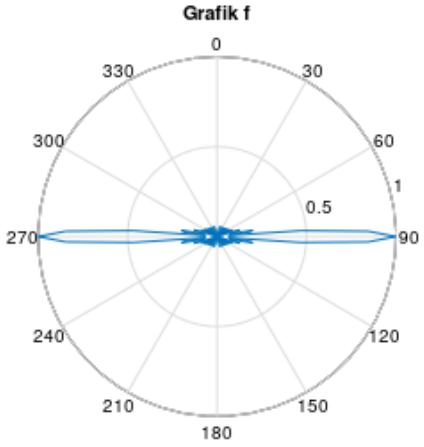


(d) n = 16

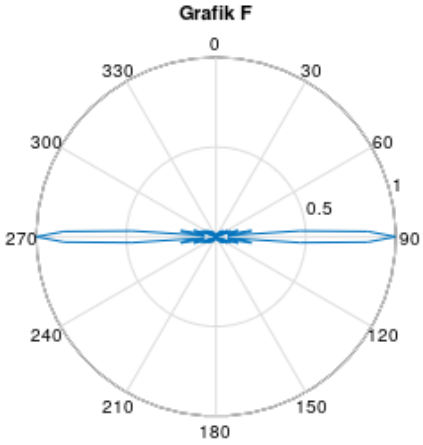
- **Pola elemen g(θ,φ)**



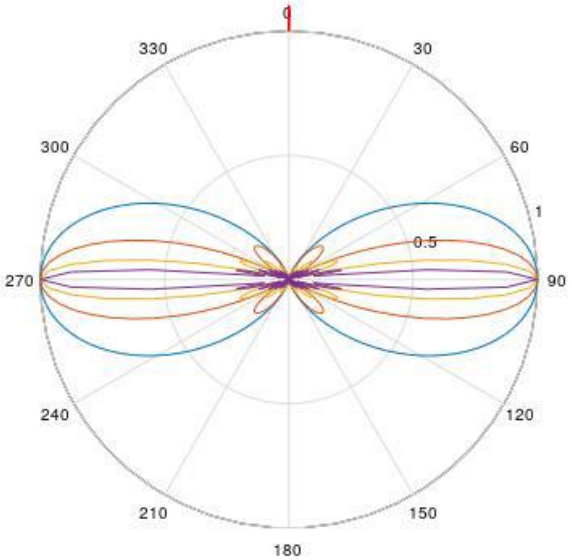
- **Pola factor susun f(θ,φ)**



- **Pola total F(θ,φ)**



- **Gambar total plot dalam satu grafik**

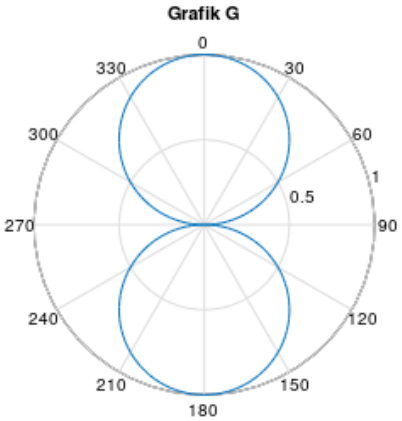


- (Blue) n = 2
- (Red) n = 4
- (Yellow) n = 6
- (Purple) n = 16

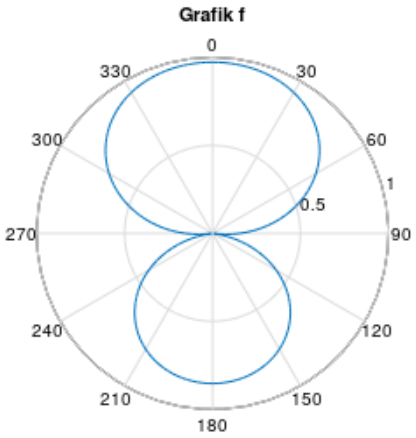
4. Sebuah antena susun (array) *endfire* terdiri dari n buah elemen antena dipol ideal. Antena susun ini diarahkan ke $\theta = 0°$. (Tetapkan pola elemennya adalah $g(\theta, \phi) = \cos\theta$). Plot pola faktor susun $f(\theta, \phi)$ dan pola total $F(\theta, \phi)$ masing-masing untuk jumlah elemen n:

(a) $n = 2$

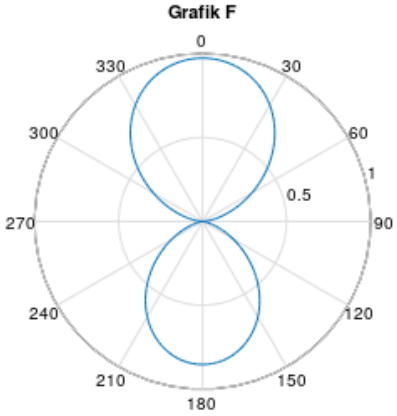
- **Pola elemen g(θ,φ)**



- **Pola factor susun f(θ,φ)**

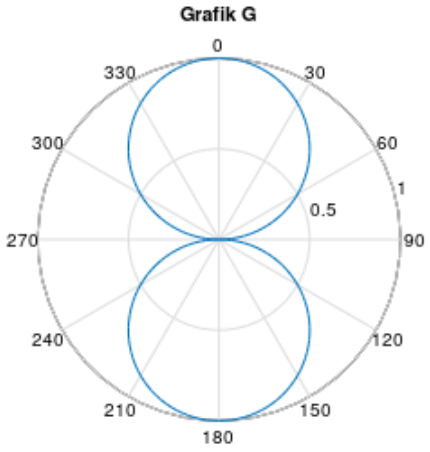


- **Pola total F(θ,φ)**

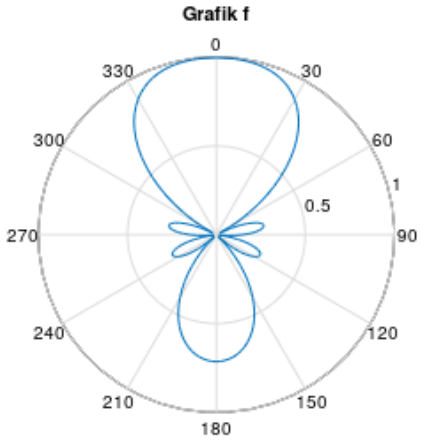


(b) $n = 4$

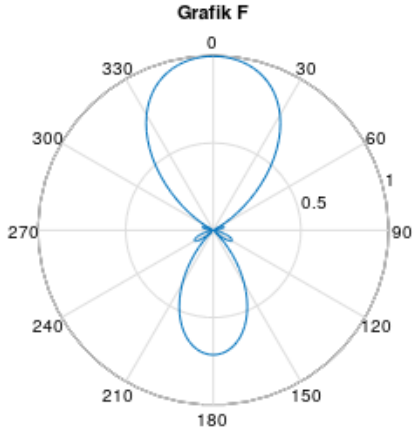
- **Pola elemen g(θ,φ)**



- **Pola factor susun f(θ,φ)**

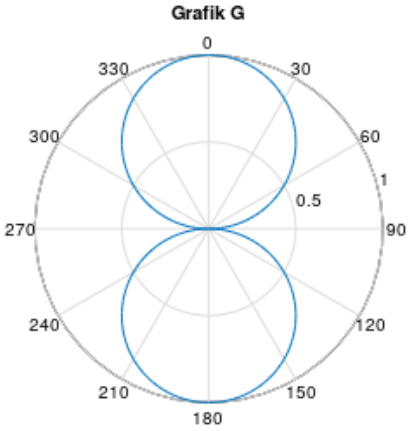


- **Pola total F(θ,φ)**

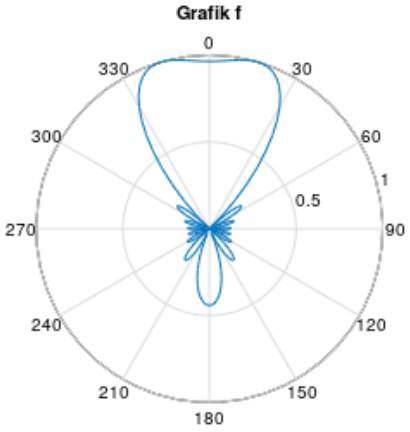


(c) n **= 8**

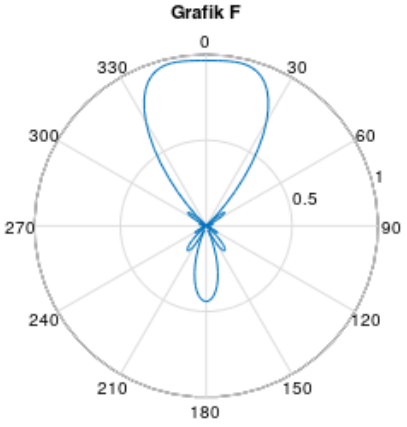
- **Pola elemen g(θ,φ)**



- **Pola factor susun f(θ,φ)**

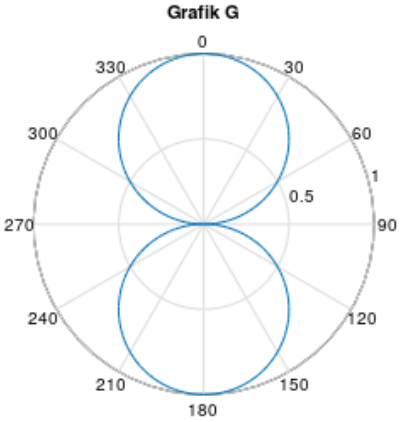


- **Pola total F(θ,φ)**

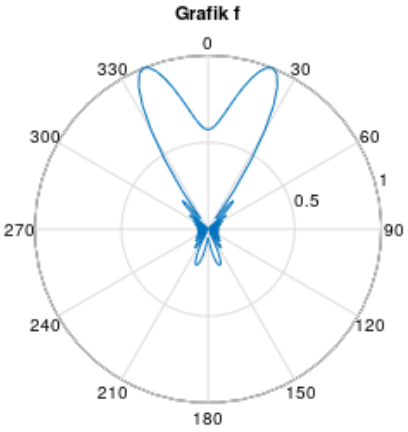


(d) n = 16

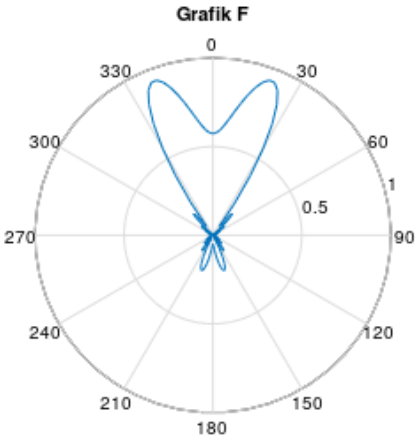
- **Pola elemen g(θ,φ)**



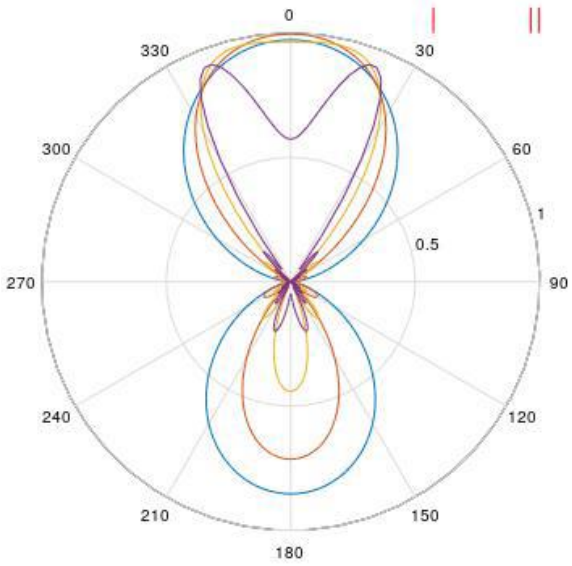
- **Pola factor susun f(θ,φ)**



- **Pola total F(θ,φ)**



- **Gambar total plot dalam satu grafik**



- (Blue) n = 2
- (Red) n = 4
- (Yellow) n = 6
- (Purple) n = 16